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Class	4
Full Name	Stephen Michael Lee
University Email	SLEE155@e.ntu.edu.sg

** : Delete and replace as appropriate.*

Declaration of Academic Integrity

By submitting this assignment for assessment, I declare that this submission is my own work, unless otherwise quoted, cited, referenced or credited. I have read and understood the Instructions to CBA.pdf provided and the Academic Integrity Policy.

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[X] I have read and accept the above.

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Declaration on Use of GenAI (Generative Artificial Intelligence)[1]

I _____Stephen Michael Lee_____ (student name),

_____SLEE155_____@e.ntu.edu.sg (NTU email) honestly and sincerely make the following declaration in relation to this submission.

In relation to the foregoing, I hereby declare that fully and properly in accordance with the Assignment/Project Instructions, I have **(insert an "X" in the square bracket if it applies)**:

i. Used GenAI as permitted to assist in generating ideas. **[ ]**

ii. Used GenAI as permitted to assist in generating answers. **[ ]**

iii. Used GenAI to refine grammar and/or correct spelling. **[ ]**

iv. Used GenAI to generate code. **[ ]**

v. Did not use GenAI in any way. **[X]**

I also declare that I have:

a. Fully and honestly submitted the digital paper trail required under the assignment instructions in the appendix of this document, and that

b. For GenAI outputs used in **(i) or (ii)**, highlighted the outputs that you disagreed or modified in your answers and briefly explain why in the Appendix; and that

c. Wherever GenAI assistance has been employed in the submission in word or paraphrase or inclusion of a significant idea or fact suggested by the GenAI, I have acknowledged this by a footnote or in-text reference in my answers; and that,

d. Apart from the foregoing notices, the submission is wholly my own work.

Stephen Michael Lee 29/03/26

…… ……..

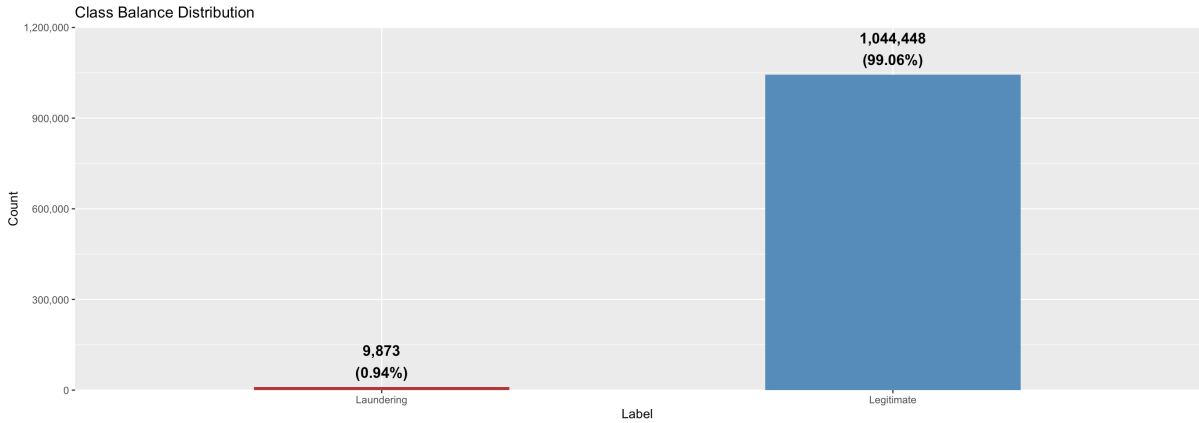
Student Name and Signature Date

[1] GenAI cannot be used to produce the entire submission. Even with GenAI, internet research or discussions, the student must refine the work on his/her own so that the submission is substantively the student's work. The list of all the prompts given to the GenAI and the GenAI outputs must be listed in the appendix of this document.

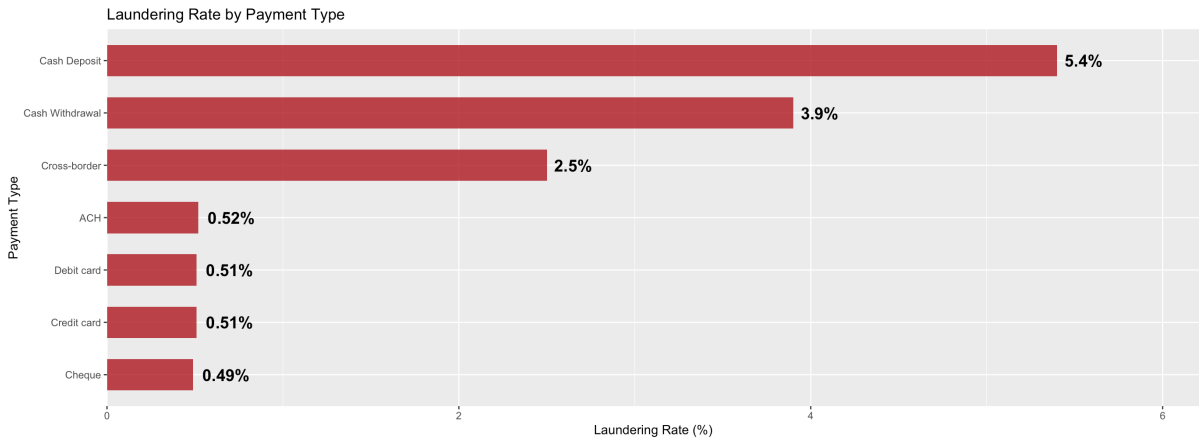
Answer to Q1:

Based on initial data checks, the dataset contains 1054321 banking transactions recorded across 321 days from 7 October 2022 to 23 August 2023, with 12 variables. No missing values or duplicate rows were found. 6 key findings are highlighted below while I conducted my Exploratory Data Analysis (EDA).

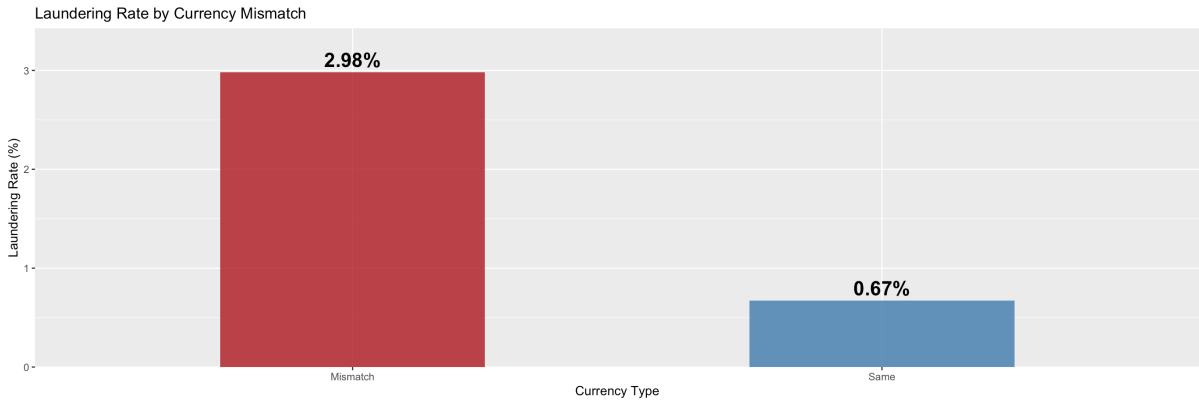
1. The dataset is highly imbalanced, with only approximately 0.94% of transactions classified as laundering. This indicates that laundering cases are rare events, which may bias models toward predicting the majority class (legitimate) and will require appropriate handling prior to model training.



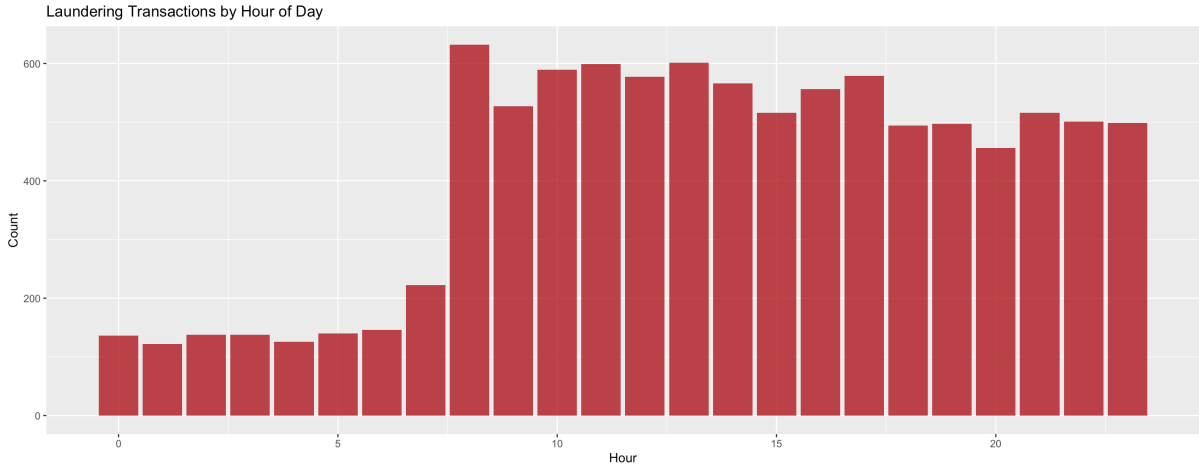
2. Cash deposits (5.4%), cash withdrawals (3.9%) and cross-border transactions (2.5%) exhibit higher laundering rates compared to other payment types. These channels are harder to trace and more susceptible to layering techniques, making them preferred modes for money laundering.



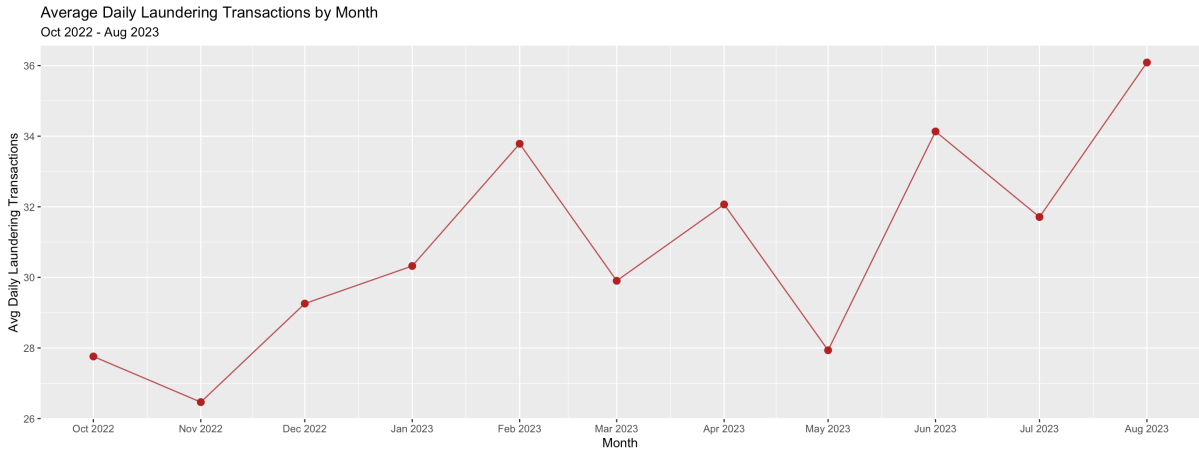
3. The laundering rate for transactions involving currency mismatch is > 4 times higher compared to same-currency transactions. Cross-currency transactions are certainly harder to monitor and regulate.



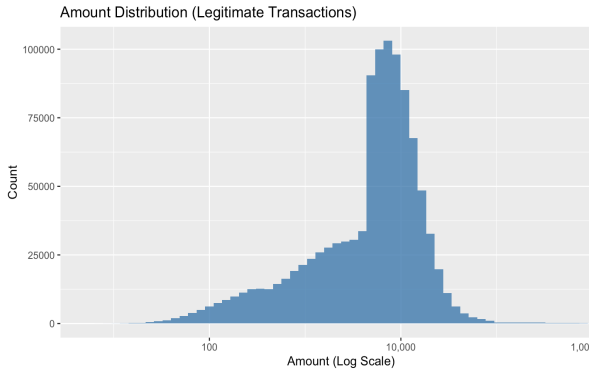
4. Laundering transactions are more frequent during typical business hours (approximately 8am to 5pm), with reduced activity during late-night periods. This suggests that illicit transactions are structured to mimic normal customer behaviour and blend into routine transaction flows.

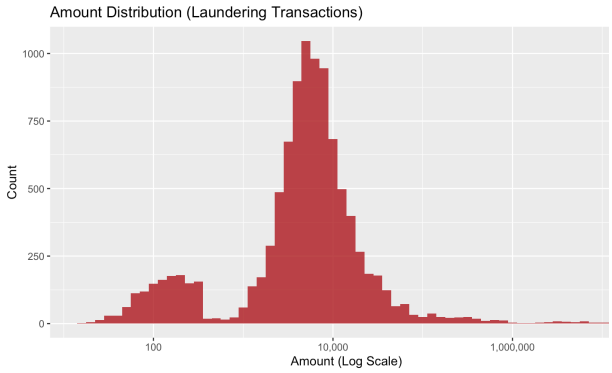


5. Average daily laundering transactions increased from approx. 26 in November 2022 to approx. 36 in August 2023, a 38.5% increase over 10 months. This growing trend underscores the need for a proactive, automated detection system rather than reactive rule-based monitoring.



6. Log transformation was used due to extreme right skew in transaction amounts. Although transaction amounts are recorded in mixed currencies, the dataset is dominated by UK-pound transactions (95.7%), so the overall shape is still influenced by UK-pound values. Legitimate transactions appear largely unimodal. Laundering transactions show a distinct bimodal distribution with a main peak alongside a secondary cluster at small amounts which is consistent with structuring/smurfing, where launderers deliberately split large sums into multiple small transactions.





Answer to Q2:

A stratified 70/30 train-test split was performed on the full dataset, resulting in 738025 observations in the training set and 316296 in the test set. Both sets maintained their original laundering proportion of 0.94%. The test set was kept unchanged throughout to reflect real-world class distribution.

```
> set.seed(2026)
> split <- sample.split(df.model$Is_laundering, SplitRatio = 0.7)
> df.train.full <- df.model[split == TRUE, ]
> df.test <- df.model[split == FALSE, ]
```

Due to computational constraints, a stratified sample of 100000 observations was drawn from the full training set and will be used later to train the models. Trade-off: Better computational efficiency but at the cost of information loss.

```
> n_sample   <- 100000
> laundering <- df.train.full[df.train.full$Is_laundering == "Laundering", ]
> legitimate <- df.train.full[df.train.full$Is_laundering == "Legitimate",  ]
> prop_l     <- nrow(laundering) / nrow(df.train.full)
> n_l        <- round(n_sample * prop_l)
> n_leg      <- n_sample - n_l
>
> df.train <- rbind(laundering[sample(nrow(laundering), n_l),  ], legitimate[sample(nrow(legitimate), n_leg), ])
> df.train <- df.train[sample(nrow(df.train)), ]
```

As missing true laundering cases carries higher risk than false alarms, the sampled training set was rebalanced using the ROSE package, combining oversampling of the minority class and undersampling of the majority class simultaneously. This resulted in approximately 59.76% laundering representation in the newly rebalanced training set, slightly above 50/50 proportion to encourage improved detection of the minority class without excessively distorting the distribution.

```
> df.train.balanced <- ovun.sample(Is_laundering ~ ., data = df.train,
+                                  N = nrow(df.train), p = 0.6, method = "both", seed = 2026)$data
>
> # Verify proportions
> prop.table(table(df.train.balanced$Is_laundering))

Legitimate Laundering
   0.40236    0.59764
> prop.table(table(df.test$Is_laundering))

 Legitimate  Laundering
0.990635354 0.009364646
```

The test set was not sampled or rebalanced, so model evaluation reflects the original imbalanced distribution where laundering transactions are rare. It was also kept completely separate throughout the entire process to avoid data leakage, which would lead to overly optimistic and less reliable performance estimates.

Observations & Proportion Table:

Dataset	Number of Observations	Number of Laundering	Proportion of Laundering (%)
Full trainset before sampling	738025	6911	0.94
Sampled trainset before balancing	100000	936	0.94
Sampled trainset after balancing	100000	59764	59.76
Full testset	316296	2962	0.94

Answer to Q3:

Before modelling, several variables were removed:

- **Laundering type:** Removed due to data leakage. It directly describes how a transaction was laundered and is only known after detection, making it unavailable at prediction time.
- **Date & Time:** Raw date and time are high cardinality. Engineered into Month and Hour.
- **Sender/Receiver bank location:** Low variance as majority is UK. Engineered into binary 'Different_bank_location' column instead to capture cross-border transactions.
- **Payment currency & Received currency:** Low variance as majority is UK. Engineered into binary 'Currency_mismatch' column instead to identify currency inconsistencies.
- **Sender account & Receiver account:** High-cardinality identifiers dropped and replaced by 5 behavioural aggregated features computed from training data only. They are 'Sender_txn_count', 'Receiver_txn_count', 'Pair_txn_count', 'Sender_avg_amount' and 'Amount_vs_sender_avg'. These capture recurring account behaviour as laundering is often pattern-based rather than identifiable from a single transaction.

```
> # Basic feature engineering
> df.clean <- df.raw %>%
+   mutate(
+     Hour = hour(Time),
+     Month = month(Date),
+     Is_laundering = factor(Is_laundering, levels = c(0, 1), labels = c("Legitimate", "Laundering")),
+     Different_bank_location = ifelse(Sender_bank_location != Receiver_bank_location, 1, 0),
+     Currency_mismatch = ifelse(Payment_currency != Received_currency, 1, 0)
+   )
>
> # Initial feature selection
> drop_cols <- c("Time", "Date", "Laundering_type", "Payment_currency",
+                "Received_currency", "Sender_bank_location", "Receiver_bank_location")
> # Attach LOO features to train set
> df.train.full <- df.train.full %>%
+   mutate(
+     Sender_txn_count = sender_loo$Sender_txn_count,
+     Sender_avg_amount = sender_loo$Sender_avg_amount,
+     Receiver_txn_count = receiver_loo$Receiver_txn_count,
+     Pair_txn_count = pair_loo$Pair_txn_count,
+     Amount_vs_sender_avg = Amount / pmax(Sender_avg_amount, 1)
+   )
> # Remove account IDs (identifiers, not predictors)
> df.train <- df.train %>% select(-Sender_account, -Receiver_account)
> df.test  <- df.test  %>% select(-Sender_account, -Receiver_account)
```

Leave-One-Out (LOO) aggregation was used so that each training row's behavioural features are computed excluding itself. All aggregates were derived solely from the training set and applied to the test set. Unseen test accounts receive a count of 0 and the training median as a safe fallback.

```
> # Leave-One-Out sender stats
> sender_loo <- df.train.full %>%
+   group_by(Sender_account) %>%
+   mutate(
+     Sender_txn_count  = n() - 1,
+     Sender_avg_amount = ifelse(
+       (n() - 1) > 0,
+       (sum(Amount) - Amount) / (n() - 1),
+       NA_real_
+     )
+   ) %>%
+   ungroup()
>
> # Fill NAs (senders with a single transaction) with the global training median
> global_median_amount <- median(df.train.full$Amount, na.rm = TRUE)
> sender_loo$Sender_avg_amount[is.na(sender_loo$Sender_avg_amount)] <- global_median_amount
```

Logistic Regression: A full baseline model was first fitted with all predictors. Multicollinearity was checked using VIF, which found 'Different_bank_location' to be abnormally high.

```
> vif(lr.m1)
                             GVIF Df GVIF^(1/(2*Df))
Amount                 1.878350e+00  1        1.370529
Payment_type           1.522517e+07  6        3.967773
Hour                   1.025150e+00  1        1.012497
Month                  1.003203e+00  1        1.001600
Different_bank_location 7.057900e+06  1     2656.670843
Currency_mismatch      4.892423e+00  1        2.211882
Sender_txn_count       1.424609e+00  1        1.193570
Sender_avg_amount      1.494800e+00  1        1.222620
Receiver_txn_count     1.472640e+00  1        1.213524
Pair_txn_count         2.479422e+00  1        1.574618
Amount_vs_sender_avg   1.583700e+00  1        1.258451
```

After removing the variable, stepwise AIC selection was then applied to remove any remaining variables that did not contribute to model fit. 'Amount' and 'Pair_txn_count' were removed.

```
> # Model 3: Step wise AIC selection
> lr.m3 <- step(lr.m2, direction = "both", trace = FALSE)
> lr.m3.prob <- predict(lr.m3, df.test, type = "response")
> lr.m3.pred <- factor(ifelse(lr.m3.prob > 0.5, "Laundering", "Legitimate"),
+                      levels = levels(df.test$Is_laundering))
> lr.m3.cm <- table(Actual = df.test$Is_laundering, Predicted = lr.m3.pred)
```

```
                Predicted
Actual        Legitimate Laundering
  Legitimate     216395      96939
  Laundering        763       2199
```

Random Forest: A baseline was first fitted with the default parameters. While this model's accuracy was high (98.46%), I wanted to improve its current sensitivity (81.87%) as it was too conservative. The classification cutoff was then adjusted to (0.6, 0.4) to lower the threshold for predicting laundering cases and number of trees were increased to 600 to stabilise predictions. Overall, sensitivity improved by 4.5% to 86.33% with accuracy traded off to 96.01%, which is still relatively high.

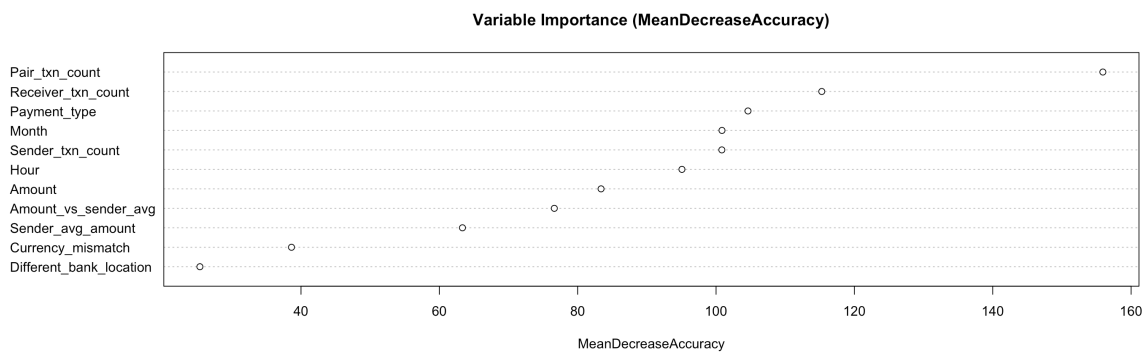
```
> # Model 3: More trees
> rf.m3 <- randomForest(Is_laundering ~ ., data = df.train.balanced,
+                       cutoff = c(0.6, 0.4), ntree = 600, na.action = na.omit, importance = TRUE)
> rf.m3.pred <- predict(rf.m3, df.test)
> rf.m3.cm   <- table(Actual = df.test$Is_laundering, Predicted = rf.m3.pred)
```

```
                Predicted
Actual        Legitimate Laundering
  Legitimate     301109      12225
  Laundering        405       2557
```

Model Results Table:

Model	Complexity	Test set Error (%)	Accuracy (%)	Sensitivity (%)	Specificity (%)
Random Forest	ntree=600, mtry=3, cutoff=(0.6, 0.4)	3.99	96.01	86.33	96.10
Logistic Regression	8 X variables	30.89	69.11	74.24	69.06

Variable Importance Plot (Random Forest):



Memo to CEO (Key Findings from Models):

The Random Forest (RF) model significantly outperformed Logistic Regression (LR) across all metrics. It correctly identified approximately 86 out of every 100 laundering transactions (sensitivity: 86.33%) while correctly clearing approximately 96 out of every 100 legitimate transactions (specificity: 96.10%). LR, while useful as a baseline model, captured only 74 in 100 laundering cases with far lower overall accuracy (69.11% vs 96.01%). This indicates that laundering behaviour is primarily driven by complex, non-linear patterns, which are better captured by RF.

The most important signals for detecting laundering are repeated transactions between the same sender-receiver pair (Pair_txn_count), high receiver transaction frequency (Receiver_txn_count), high sender transaction frequency (Sender_txn_count), and payment type. For payment type, particularly cash deposits and withdrawals which showed the highest laundering rates during the earlier EDA. The RF model is recommended to be deployed as a first-line screening tool, with all flagged transactions reviewed by compliance officers before action is taken. This balances strong detection performance against unnecessary disruption to legitimate customers.

Answer to Q4:

I partially disagree. While grid search systematically tests parameter combinations to find a better-performing configuration, it is not strictly necessary to prevent overfitting in Random Forest.

Random Forest is already less prone to overfitting due to its ensemble design. Bagging reduces variance and random feature selection at each split decorrelates the trees, making the overall model more robust. In practice, Random Forest often performs well even with just its default parameters.

However, poor parameter choices (overly complex trees or insufficient randomness) can still lead to overfitting. Grid search may then be useful for improving performance, but it should not be viewed as the only or necessary way to control overfitting. More importantly, tuning must be combined with proper validation, such as cross-validation or evaluation on an unseen test set. If grid search is performed only on the training data without adequate validation, it may itself lead to overfitting.

Additionally, grid search can be computationally intensive as it evaluates every possible parameter combination, especially for large datasets with multiple hyperparameters. Hence, simpler tuning approaches may be more practical.

Overall, grid search is helpful for optimisation, but good validation practice is more important than grid search alone to prevent overfitting.

[198 words]

Answer to Q5:

To improve the success of analytics in anti-money laundering, banks should focus on both data and implementation.

1. Engineer richer features:

Include data involving transaction history, network relationships between accounts and customer profiles to better capture behavioural patterns. Network graph features such as transaction velocity across linked accounts help detect coordinated laundering rings, where multiple accounts work together in ways invisible at the individual transaction level.

2. Feedback loop and continuous model monitoring:

A model that performs well today may underperform in the future as criminal behaviour adapts. Structured processes should be established, where compliance officers log investigation outcomes and are then fed back periodically to retrain the model. This allows it to adapt as laundering strategies evolve over time. Implement automated monitoring to detect when model performance degrades, triggering scheduled retraining.

3. Explainability for operational adoption:

Deploy model explanations alongside predictions so compliance officers understand why a transaction is flagged. This improves triage speed, builds officer trust in the model and enables more effective investigation prioritisation.

4. Balance detection with customer experience:

Detection thresholds should be carefully finetuned to maximise detection while minimising unnecessary disruption to legitimate customers. Overly aggressive flagging risks disrupting legitimate customers and undermining business objectives.

[200 words]

Appendix: List of GenAI Prompts and Outputs

[For each question in the paper, state (a) the complete list of GenAI prompts used (if any), in (b) which GenAI tool (ChatGPT, Gemini, etc…) and (c) entire GenAI Outputs under each GenAI prompt. In addition, for GenAI used as in (i) or (ii), highlight the outputs that you disagreed or modified in your answers and briefly explain why. Marks will be deducted for non-compliance. If you are unsure, ask your instructor.]